

# Ninad Alurkar

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## Summary

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Robotics and AI Engineer specializing in autonomous systems, robot perception, and human-robot interaction. Experience building end-to-end ROS-based pipelines integrating SLAM, sensor fusion, and deep learning for real-world robotic platforms. Strong background in reinforcement learning, computer vision, and multimodal interaction systems, with demonstrated improvements in task performance and system efficiency.

## Education

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### University of Michigan–Dearborn

Sep 2024 – Dec 2025

Master of Science in Robotics Engineering (High Distinction)

GPA: 3.96 / 4.00

Relevant Coursework: Vehicle Mobility Systems, Robotics Perception, Advanced Control Systems

### National Institute of Technology–Surat

Jun 2018 – Apr 2022

Bachelor of Technology in Mechanical Engineering

CGPA: 8.60 / 10.00

## Technical Skills

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**Programming:** C++, Python, MATLAB, Simulink

**ADAS / Robotics:** ROS/ROS2, SLAM, sensor fusion, MoveIt

**Simulation & Validation:** Gazebo, RViz, MuJoCo (CARLA-transferable), closed-loop testing

**Machine Learning:** PyTorch, TensorFlow, Deep Learning, RL (PPO), transformers

**Perception & Systems:** OpenCV, CUDA, Linux, SolidWorks, ANSYS

## Professional Experience

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### Software Engineer Intern

Feb 2026 – May 2026

OpenQQuantify, Falls Church, VA (Remote)

- Contributed to the development of AI-enabled media platforms and software systems supporting AI-driven electronic design.
- Implemented AI/ML-based components and assisted in the development and optimization of AI-integrated electronic systems.

### R&D Engineer

Jan 2025 – Aug 2025

Electrical and Computer Engineering Department, University of Michigan–Dearborn

- Led development of an autonomous mobile robot platform (Clearpath JACKAL) for greenhouse operations, architecting ROS-based pipelines for LiDAR SLAM, perception, and navigation.
- Designed and deployed a multi-sensor perception stack integrating LiDAR and vision, enabling environment mapping and object classification in simulation and real-world settings.
- Developed and validated computer vision and machine learning models achieving 98% classification accuracy, evaluated in Gazebo and RViz.
- Directed an engineering team, defining system architecture, integration workflows, and testing protocols for end-to-end robotic deployment.

### Software Engineer

Aug 2022 – Aug 2024

Dover Corporation, Bengaluru, India

- Developed Python- and SQL-based data processing and validation pipelines, reducing manual engineering review effort by 40%.
- Collaborated with mechanical and systems engineers to support sensor-data simulation workflows for system validation.

## Research and Projects

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### Reinforcement Learning-Based Control in MuJoCo Simulation

Jan 2026

- Developed a mobile-robot autonomy environment in MuJoCo and trained a PPO controller, improving task success from 28% to 71%.

- Reduced average time-to-goal by 62% (8.7 s to 3.3 s) versus a rule-based baseline, demonstrating performance gains under closed-loop control.

### **ML-Based Ridge Regression (L2) Strategy for Time-Series Prediction**

Jan 2026

- Designed a rolling-window Ridge Regression pipeline that reduced out-of-sample error by 15% and improved robustness under covariate shift.

### **Trust Building in Human-Robot Interaction (Master's Thesis)**

Jan 2024 – Jan 2026

- Designed and implemented a companion robotic system with adaptive multimodal interaction, building a full ROS architecture on a Clearpath JACKAL integrating an OAK-D Pro camera and ReSpeaker microphone array.
- Integrated vision and audio into a synchronized perception pipeline operating in real-time.
- Conducted an IRB-approved human study with 60 participants, analyzing trust and system reliability using Cronbach's alpha and statistical performance metrics.

### **Temporal Prediction Models: LSTM vs. Transformer**

Sep 2025 – Dec 2025

- Built GPU-accelerated forecasting pipelines in PyTorch and TensorFlow, benchmarking models using RMSE and MAE across varying volatility regimes.

### **Autonomous Manipulation Using Computer Vision**

Jan 2025 – Mar 2025

- Designed and simulated a closed-loop perception–planning–control system in ROS and Gazebo using OpenCV-based vision, reducing manual task time by 7 minutes.